

Smart charging of electric vehicles



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Managerial motivation



Introduction: Managerial motivation

- ▶ **Problem class:** LP / convex QP
- ▶ We look at a new operational problem, created by the energy transition
- ▶ We will understand why a decision that *seems* to be of on/off type is in fact modelled with continuous variables

The fleet problem

- ▶ A company fleet must be fully charged by the morning by exploiting the cheap night hours
- ▶ But if all the vehicles charge together during the cheap hours, the **peak draw** explodes
- ▶ With it explode the power costs and the risk of exceeding the capacity of the meter
- ▶ The model decides *how much power* to give to each vehicle in each hour: a naturally **continuous** decision

Smart charging

- ▶ Smart charging is one of the new operational problems created by the energy transition
- ▶ Ingredients: electric fleets, hourly energy prices, power limits of the meters
- ▶ A perfect example of a problem that *seems* to require on/off variables and instead is a **pure LP**
- ▶ The reason: the real decision — how much power — is continuous

In this chapter we will learn how to

1. model decisions over an hourly horizon with availability windows
2. read the duals as “prices of the marginal hours”
3. handle the cost-peak trade-off with a multi-objective approach
4. diagnose a real pitfall of sensitivity analysis (the right-hand side with constants)

Summary

- ▶ Goal: charge the fleet at night at minimum cost
- ▶ Risk: concentrating all the charging in the cheap hours creates a peak of draw
- ▶ The decision is **how much power** per vehicle and per hour: continuous variables, no binary variables

Guided construction of the model

Introduction: Guided construction of the model

- ▶ We describe the problem in words, then fix data and variables
- ▶ We write the minimum-cost charging LP model and explain every family of constraints
- ▶ Finally we see the variants: peak shaving, smooth profile, multi-objective

The problem in words

- ▶ *We decide* how much power to deliver to each vehicle in each hour
- ▶ *The objective* is to minimize the energy expenditure of the night
- ▶ *The constraints*:
 - ▶ every vehicle must receive all the energy it needs before departure
 - ▶ it can charge only while it is plugged in and never above the power of its charger
 - ▶ the overall draw (charging plus building) can never exceed the power of the meter

Data (input of the model) (1/2)

Symbol	Type	Meaning
n	$\in \mathbb{Z}_{\geq 1}$	number of vehicles, indexed by $v \in \{1, 2, \dots, n\}$
m	$\in \mathbb{Z}_{\geq 1}$	number of hourly intervals, indexed by $t \in \{1, 2, \dots, m\}$, of duration Δt (here $\Delta t = 1$ h)
π_t	$\in \mathbb{Q}_{\geq 0}$	energy price in hour t (€/kWh)
a_{vt}	$\in \{0, 1\}$	observed <i>datum</i> : 1 if vehicle v is plugged in during hour t , 0 otherwise
e_v	$\in \mathbb{Q}_{> 0}$	energy required by vehicle v before departure (kWh)

Data (input of the model) (2/2)

Symbol	Type	Meaning
$\bar{p}_v$	$\in \mathbb{Q}_{>0}$	maximum charging power of vehicle v (kW)
b_t	$\in \mathbb{Q}_{\geq 0}$	base load of the building in hour t (kW)
k	$\in \mathbb{Q}_{>0}$	maximum power that can be drawn from the meter (kW)
η	$\in \mathbb{Q}, 0 < \eta \leq 1$	charging efficiency

Decision variables

We introduce the following $n m$ non-negative variables:

x_{vt} = power assigned to vehicle v in hour t (kW),

$$\forall v \in \{1, 2, \dots, n\}, \forall t \in \{1, 2, \dots, m\}.$$

Using these variables, we now write an LP model for the problem.

Minimum-cost charging LP model

Minimum-cost charging LP model

$$\min \sum_{v=1}^n \sum_{t=1}^m \pi_t \Delta t x_{vt} \quad (10.1a)$$

$$\text{subject to } \eta \sum_{t=1}^m \Delta t x_{vt} \geq e_v, \quad \forall v \in \{1, 2, \dots, n\}, \quad (10.1b)$$

$$\sum_{v=1}^n x_{vt} + b_t \leq k, \quad \forall t \in \{1, 2, \dots, m\}, \quad (10.1c)$$

$$x_{vt} \leq a_{vt} \bar{p}_v, \quad \forall v \in \{1, 2, \dots, n\}, \quad \forall t \in \{1, 2, \dots, m\}, \quad (10.1d)$$

$$x_{vt} \geq 0, \quad \forall v \in \{1, 2, \dots, n\}, \quad \forall t \in \{1, 2, \dots, m\}. \quad (10.1e)$$

Objective function and energy constraints

- ▶ The linear objective function minimizes the energy expenditure of the night: the power x_{vt} delivered for the duration Δt costs $\pi_t \Delta t x_{vt}$ euros
- ▶ The energy constraints ensure that every vehicle receives all the energy it requires before departure (n linear constraints)
- ▶ The efficiency η stands on the left: of the energy that is drawn, only the fraction η ends up in the battery

Grid constraints, power constraints and domain

- ▶ The grid constraints impose that the overall draw (charging plus base load) never exceeds the power of the meter: they **couple the vehicles** to each other (m linear constraints)
- ▶ The power constraints limit every vehicle to its charger and set the power to zero in the hours in which the vehicle is absent ($a_{vt} = 0$) ($n m$ linear constraints)
- ▶ Availability is a *datum*, not a decision: this is why **no binary variables are needed**
- ▶ The constraints $x_{vt} \geq 0$ define the variables of the model

Variants of the model

- ▶ **Peak shaving:** $\min z$ with

$$\sum_{v=1}^n x_{vt} + b_t \leq z, \quad \forall t \in \{1, 2, \dots, m\}$$

- ▶ **Smooth profile (QP):** term $\gamma \sum_{t=2}^m (z_t - z_{t-1})^2$
- ▶ **Multi-objective:** $\text{cost} + \rho z$

The energy constraint written as $\geq$

- ▶ Charging more than necessary is allowed
- ▶ But it is never worthwhile: the price is positive
- ▶ Hence at the optimum the energy constraint holds **with equality**

Summary

- ▶ Continuous variables x_{vt} : power per vehicle and per hour
- ▶ Objective: energy expenditure $\sum_v \sum_t \pi_t \Delta t x_{vt}$
- ▶ Constraints: energy per vehicle, meter capacity per hour, charger power and availability
- ▶ Availability a_{vt} is a datum: the model remains a pure LP

Worked example by hand

Introduction: Worked example by hand

- ▶ We solve by hand a minimal case: one vehicle, two hours
- ▶ We compute the optimal solution and the shadow prices
- ▶ We obtain the key for reading the whole chapter: the dual is the price of the marginal hour

Example: one vehicle, two hours (1/3)

- ▶ We need $e = 10$ kWh ($\eta = 1$)
- ▶ Prices: $\pi_1 = 0.10$ and $\pi_2 = 0.20$ €/kWh
- ▶ Maximum power: $\bar{p} = 8$ kW

Example: one vehicle, two hours (2/3)

Case $\bar{p} = 8$.

- ▶ In the cheap hour we charge at the maximum (8 kWh); the remaining 2 kWh go into the expensive hour
- ▶ Cost: $8 \cdot 0.10 + 2 \cdot 0.20 = 1.20$ €
- ▶ *Shadow price of the energy requirement*: the eleventh kWh would end up in the expensive hour $\Rightarrow 0.20$ €/kWh
- ▶ *Shadow price of the power limit in hour 1*: one more kW would allow us to move one kWh from the expensive hour to the cheap one $\Rightarrow 0.10$ €/kW

Example: one vehicle, two hours (3/3)

Case $\bar{p} = 12$.

- ▶ Everything in the cheap hour: cost 1.00 €
- ▶ Dual of the requirement: 0.10
- ▶ Dual of the power: 0 (constraint not active)

The pattern “**the dual is the price of the marginal hour**” is the key for reading the whole chapter.

Summary

- ▶ The cheap hours are filled first, up to the power limit
- ▶ The shadow price of the energy requirement is the price of the marginal hour
- ▶ The shadow price of the power limit is the saving obtained by moving energy from the expensive hour to the cheap one; it is zero if the constraint is not active

Case study

Introduction: Case study

- ▶ We apply the model to a realistic fleet
- ▶ We see how the model is written in Python with Gurobi
- ▶ We get ready to read the results and the shadow prices

The data of the case study

- ▶ Six vans with different night charging windows (data in `dati/ev_flotta.csv`)
- ▶ Stylized hourly prices with an evening peak
- ▶ Base load of the building
- ▶ A meter of $k = 120$ kW, efficiency $\eta = 0.95$

The model in Gurobi (1/2)

Variables, with the power limit and the availability as upper bound:

```
x = m.addVars(veicoli, ore, name="x")
for v in veicoli:
    for t in ore:
        # 0 if not plugged in
        x[v, t].UB = P[v] * disp[v, t]
```

The model in Gurobi (2/2)

Energy and grid constraints, objective function:

```
v_ene = m.addConstrs((eta * gp.quicksum(x[v, t]
                                         for t in ore)
                      >= E[v] for v in veicoli),
                     name="energia")
v_rete = m.addConstrs((gp.quicksum(x[v, t]
                                   for v in veicoli)
                      + base[t] <= C for t in ore),
                     name="rete")
m.setObjective(gp.quicksum(prezzo[t] * x[v, t]
                           for v in veicoli for t in ore),
              GRB.MINIMIZE)
```

Summary

- ▶ Six vans, different windows, a meter of 120 kW, $\eta = 0.95$
- ▶ The power constraint simply becomes the upper bound of the variable (0 if the vehicle is not plugged in)
- ▶ Energy and grid are two families of constraints with `addConstrs`

Results

Introduction: Results

- ▶ We compare the minimum-cost strategy with peak shaving
- ▶ We look at the draw profiles hour by hour
- ▶ We interpret the shadow prices of the energy requirement as prices of the marginal hours

Output of the model

Minimum cost : 20.36 EUR/night

peak draw 103.4 kW (limit 120)

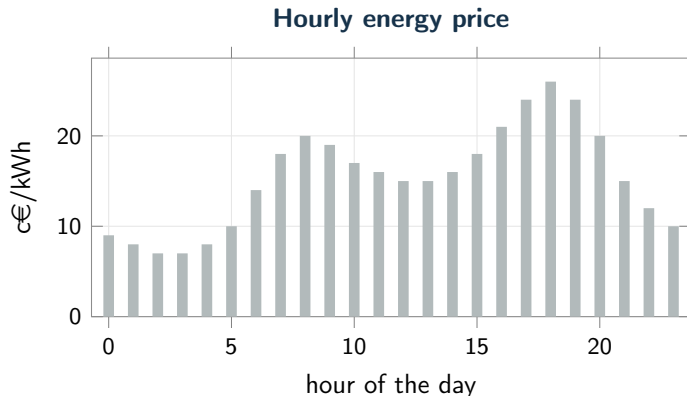
Peak shaving : minimum possible peak 68.0 kW

Shadow prices of the energy requirement (EUR/kWh):

V1	0.0947	V2	0.0842	V3	0.0842
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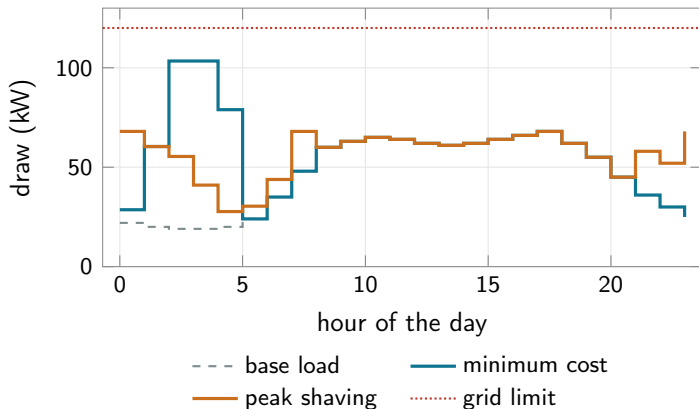
V4	0.0947	V5	0.0947	V6	0.0842
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Hourly energy price



The price is low at night and has an evening peak.

Total draw: chasing the prices creates a night-time peak



Reading the chart

- ▶ The chart shows the total draw (base + charging) hour by hour under the two strategies
- ▶ The **minimum cost** one concentrates the charging in the cheap hours and creates a night-time peak of 103 kW
- ▶ **Peak shaving** spreads the charging out and keeps the draw at 68 kW
- ▶ Both stay below the **grid limit** of 120 kW

The duals tell the story of the marginal hours (1/2)

- ▶ The shadow prices of the energy requirement are equal to

$$0.0842 = 0.08/\eta \quad \text{or} \quad 0.0947 = 0.09/\eta$$

- ▶ They are the prices of the *marginal hours* of each vehicle (the cheapest hour still free within its window), corrected for the efficiency

The duals tell the story of the marginal hours (2/2)

- ▶ V2, V3 and V6 have windows that cover the hours at 0.08
- ▶ V1, V4 and V5 arrive late or leave early: their marginal hour costs 0.09
- ▶ One additional kWh of requirement therefore costs between 8.4 and 9.5 cents depending on the vehicle

Summary

- ▶ Minimum cost: 20.36 €/night with a peak of 103.4 kW
- ▶ Peak shaving: minimum possible peak 68 kW
- ▶ Shadow prices of the requirement = price of the marginal hour/ η : 8.4 or 9.5 cents per kWh

Sensitivity analysis

Introduction: Sensitivity analysis

- ▶ We study the trade-off between cost and peak by varying the weight ρ of the peak
- ▶ We vary the capacity k of the meter
- ▶ We see an instructive bug, which really happened, in the modification of the right-hand side

Trade-off cost + $\rho \cdot \text{peak}$

Trade-off cost + $\rho \cdot \text{peak}$:

rho = 0.00:	cost	20.36	peak	103.4
rho = 0.05:	cost	20.69	peak	86.5
rho = 0.10:	cost	21.39	peak	74.9
rho = 0.20:	cost	22.15	peak	68.0
rho = 0.50:	cost	22.15	peak	68.0

Meter capacity

Meter capacity:

k = 60, 65 kW: INFEASIBLE

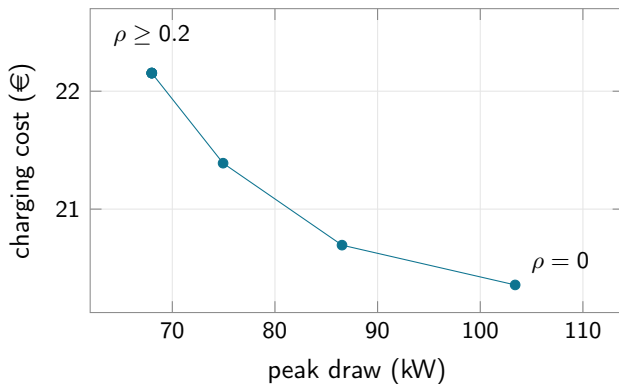
k = 70 kW : 21.93

k = 80 kW : 21.09

k = 90 kW : 20.63

k = 120 kW : 20.36

Cost-peak frontier



Reading the frontier

- ▶ The frontier is obtained by varying the weight ρ of the peak in the objective
- ▶ Every point is an optimal charging strategy
- ▶ Reducing the peak from 103 to 75 kW costs about one euro per night
- ▶ The point $\rho \geq 0.2$ reaches the minimum possible peak (68 kW) at the lowest cost that allows it

Peak shaving pays for itself

- ▶ Cutting the peak from 103 to 75 kW (-28%) costs only one euro per night; getting down to 68 kW costs less than two
- ▶ If the power tariff or the upgrade of the meter cost more than that, peak shaving pays for itself
- ▶ The pure minimax (peak 68) without the cost term would spend 27.79 €
- ▶ The trade-off with $\rho = 0.2$ achieves *the same peak* at 22.15 €:
never optimize a single objective when there are two of them

An instructive bug, which really happened (1/2)

Observation 1: An instructive bug, which really happened

In the constraint $\text{quicksum}(x) + \text{base}[t] \leq C$, Gurobi moves the constant $\text{base}[t]$ into the right-hand side: the stored RHS is $C - \text{base}[t]$.

The first version of our sensitivity analysis wrote $v.RHS = \text{nuovaC}$ and obtained identical costs for every capacity.

An instructive bug, which really happened (2/2)

Observation 2: An instructive bug, which really happened

It was in fact relaxing the constraint by $\text{base}[t]$.

The correct form is $v.RHS = \text{nuovaC} - \text{base}[t]$.

When a sensitivity analysis changes nothing, suspect your own code before suspecting the model.

Summary

- ▶ The weight ρ traces the cost-peak frontier: with about one euro per night the peak drops from 103 to 75 kW
- ▶ Below 70 kW of meter capacity the problem becomes infeasible
- ▶ The constants in the constraints end up in the RHS: modify it taking them into account

Exercises

Introduction: Exercises

- ▶ Five exercises to put the model to the test
- ▶ Checking the duals, changes in the windows, QP variants, emissions, minimum capacity
- ▶ They are solved by modifying the Python script of the chapter

Exercise: check

Verify by perturbation the shadow price of the energy requirement of V1:

- ▶ raise e_1 from 46 to 47 kWh
- ▶ explain why it is equal to 0.09/0.95

Exercise: late arrival

V3 arrives at 23 instead of 20:

- ▶ how do cost and peak change?
- ▶ which shadow price increases?

Exercise: smooth profile

Implement the QP variant with

$$\gamma \sum_{t=2}^m (z_t - z_{t-1})^2, \quad z_t = b_t + \sum_{v \in V} x_{vt},$$

for $\gamma \in \{0.01; 0.1; 1\}$.

Compare cost, peak and “jumpiness” of the profile.

Exercise: emissions

- ▶ With an hourly carbon intensity g_t (high in the evening, low at night), draw the cost-emissions frontier
- ▶ Are the cheap hours also the cleanest ones in this case?

Exercise: minimum capacity

- ▶ Find by bisection the minimum capacity k that keeps the problem feasible
- ▶ Explain which combination of windows and $\bar{p}_v$ determines it

Summary (1/2)

Model:

- ▶ Smart charging is a pure LP: the decision is how much power, availability is a datum
- ▶ Energy constraints, grid constraints (they couple the vehicles) and power constraints

Duals:

- ▶ The shadow price of the requirement is the price of the marginal hour, divided by η

Summary (2/2)

Multi-objective:

- ▶ Cost + ρ peak traces the frontier; the trade-off reaches the minimum peak at a cost much lower than the pure minimax

Sensitivity:

- ▶ The constants of the constraints end up in the RHS: if a sensitivity analysis changes nothing, check the code first